Curriculum Vitae

1) Personal information

Name: Yunan Hu, M.S.

Email: yunan@ad.unc.edu Personal website: https://yunan0525.github.io/

2) Education

Degree	Institution	Date conferred	Major
Ph. D.	University of North Carolina at Chapel Hill (UNC-CH), USA	Present	Microbiology and Immunology
M.S.	University of North Carolina at Chapel Hill (UNC-CH), USA	May 2023	Nutrition
B.S.	China Agricultural University (CAU), China	May 2021	Food Science and Engineering

3) Bibliography and products of scholarship

Peer-reviewed publications

Career Total = 10 Google Scholar h-index = 7 Google Scholar citations=309

*1. **Y Hu,** A Rueppell, N Devpura, R Zasloff, J Arnold, J Roach, S Magness, J Bruno-Bárcena, MA Azcarate-Peril. Prebiotic β (1-4) Galacto-oligosaccharides Strengthen Intestinal Barrier Integrity via Mucus Production and Modulation of Tight Junction Genes. Food Bioscience. 2025,108052. DOI:10.1016/j.fbio.2025.108052*

*2. MR Aljumaah, J Roach, **Y Hu**, J Gunstad, MA Azcarate-Peril. Microbial dipeptidyl peptidases of the S9B family as host-microbe isozymes. Sci Adv. 2025 Apr 4;11(14):eads5721. doi: 10.1126/sciadv.ads5721.*

*3. **Y Hu**, MR Aljumaah, MA Azcarate-Peril. Galacto-Oligosaccharides and the Elderly Gut: Implications for Immune Restoration and Health. Advances in Nutrition 2024;15(8):100263. doi: https://doi.org/10.1016/j.advnut.2024.100263.*

*4. J Kang, R Li, **Y Hu**, X Huang, X-X Chen, B-Z Han. Microbial interactions in mixed-species biofilms on the surfaces of Baijiu brewing environments. Food Research International 2024;191:114698. doi: https://doi.org/10.1016/j.foodres.2024.114698.*

*5. S Tian, **Y Hu**, X Chen, C Liu, Y Xue, B Han. Green synthesis of silver nanoparticles using sodium alginate and tannic acid: characterization and anti-S. aureus activity. International Journal of Biological Macromolecules 2022;195:515-22. doi: https://doi.org/10.1016/j.ijbiomac.2021.12.031.*

*6. J Kang, **Y Hu**, L Jia, M Zhang, Z Zhang, X Huang, X Chen, B-Z Han. Response of microbial community assembly and succession pattern to abiotic factors during the second*

round of light-flavor Baijiu fermentation. Food Research International 2022;162:111915. doi: https://doi.org/10.1016/j.foodres.2022.111915.

*7. J Kang, **Y Hu**, Z Ding, L Ye, H Li, J Cheng, L Fan, H Zhao, B Han, X Zheng. Deciphering the Shifts in Microbial Community Diversity From Material Pretreatment to Saccharification Process of Fuyu-Flavor Baijiu. Front Microbiol 2021;12:705967. doi: 10.3389/fmicb.2021.705967.*

*8. **Y Hu**, X Huang, B Yang, X Zhang, Y Han, X-X Chen, B-Z Han. Contrasting the microbial community and metabolic profile of three types of light-flavor Daqu. Food Bioscience 2021;44:101395. doi: https://doi.org/10.1016/j.fbio.2021.101395.*

*9. **胡雨楠**，张钟晓，陈小雪，薛岩松，韩北忠．发酵时间对清香型白酒二茬基酒风味物质组成及含量的影响．中国酿造 2021;40(9):165-72.*

*10. X Chen, **Y Hu**, S Tian, B Han. Understanding the Interactions between Staphylococcus aureus and the Raw-Meat-Processing Environment Isolate Klebsiella oxytoca in Dual-Species Biofilms via Discovering an Altered Metabolic Profile. Microorganisms 2021;9.*

Publications in process

*1. **Y Hu,** J Arnold, et al. Mechanisms of Galectin 1 Induction by Galactooligosaccharides.*

*2. **Y Hu,** MR Aljumaah, J Roach, et al. Spatial ecology of the multi-kingdom human intestinal microbiome. Doi: https://doi.org/10.21203/rs.3.rs-9927529/v1*

*3. MR Aljumaah, I Isawi, J Roach, **Y Hu**, et al. Gut Microbial Dipeptidyl Peptidase-4 and Associated Prolyl Tripeptidyl Peptidases: A Novel Classification and Response to Inhibitors. Doi: https://doi.org/10.21203/rs.3.rs-7312416/v1*

*4. BR Wang, **Y Hu**, et al. Structural characterization and hepatoprotective mechanism of alkali-soluble Poria cocos polysaccharides modified by dynamic high-pressure microfluidization.*

*5. **Y Hu**, et al. Lactobacillus co-culture dynamics in response to different carbohydrates.*

*6. **Y Hu**, M Moore, et al. Synergistic Effects of β (1-4) Galactooligosaccharides Combined with Rationally Designed Consortia on Preventing Clostridioides difficile Infection.*

Posters and conference proceedings

Career Total = 8

1. *Conference: International Scientific Association for Probiotics and Prebiotics (ISAPP) 2026, Japan:* (Poster) **Yunan Hu,** Madison Moore, Jeff Roach, Jeena James, Jason W. Arnold, Rita Tamayo, José M. Bruno-Bárcena, Andrea Azcarate-Peril. Pure β(1-4) Galacto-oligosaccharides Enable the Persistence of a Rationally Designed Consortium in the Intestinal Tract.

2. *Conference: Gordon Research Conference (GRC): Lactic Acid Bacteria Biology, Symbioses and Applications, 2025, CA*: (Oral presentation and poster) **Yunan Hu**, Madison Moore, Jeff Roach, Jeena James, Jason W. Arnold, Rita Tamayo, José M. Bruno-Bárcena, M. Andrea Azcarate-Peril1. Pure prebiotics enable the persistence of probiotics and enhance their beneficial effects.

3. *Conference: North Carolina Branch of the American Society for Microbiology (NC-ASM) 2024, NC*: (Oral presentation) **Yunan Hu**: Galacto-oligosaccharides Enhance Intestinal Barrier Function.

4. *Conference: American Society for Nutrition (ASN) 2024, Boston:* (Poster) Jeena M James, **Yunan Hu**, Mashael Aljumaah, Esmat Rousta, Nivedita Pareek, Jeff Roach, Jason Arnold, Rita Tamayo, José Bruno-Bárcena, Andrea Azcarate-Peril. The Beneficial Effects of a Combination of Prebiotics and Bacterial Consortium on Clostridioides difficile Infection in a Mouse Model. 2024, Current Developments in Nutrition.
5. *Conference: American Society for Nutrition (ASN) 2023, Boston:* (Poster) **Yunan Hu**, Jason Arnold, Johanna Smeekens, Michael Kulis, Ellyce San Pedro, Scott Magness, Andrea Azcarate-Peril. Immunomodulatory Effects of Galactooligosaccharides. 2023, Current Developments in Nutrition.
6. *Conference: International Scientific Association for Probiotics and Prebiotics (ISAPP) 2023, Denver:* (Poster) **Yunan Hu**, Jason Arnold, Johanna Smeekens, Michael Kulis, Ellyce San Pedro, Scott Magness, Andrea Azcarate-Peril. Immunomodulatory Effect of Galacto-oligosaccharide on Lgals-1 Gene Expression.
7. *NC Microbiome Consortium Annual Symposium, 2023, NC*: (Poster) **Yunan Hu**, Jason W. Arnold, Rebecca Zasloff, Anika Rueppell, Scott T. Magness, José M. Bruno-Bárcena, M. Andrea Azcarate-Peril. Galacto-oligosaccharides Enhance Intestinal Barrier Function.
8. *NORC Student Research Symposium, 2022, NC:* (Poster) **Yunan Hu**, Jason Arnold, Johanna Smeekens, Michael Kulis, Ellyce San Pedro, Scott Magness, Andrea Azcarate-Peril. Mechanism of Galectin-1 induction by Galactooligosaccharides.

4) Mentoring Activities

Undergraduate Student Supervised Research

The table below lists supervised undergraduates who gained research experience from me. The responsibilities of students include performing research on a defined project, demonstrating skills in standard scientific methods, evaluating scientific data, and scientific troubleshooting.

Student	Degree/activities	Dates	Currently at
Garrison Bruber	BS. Postbaccalaureate research	May 2026 -	UNC-Chapel Hill
Louis Lee	Biology	May 2026 -	UNC-Chapel Hill
Inhwan Choi	BBSP PhD rotation	Nov 2025 – Feb 2026	UNC-Chapel Hill
Angela Molina	Nutrition	Aug 2025-	UNC-Chapel Hill
Jin Ho Kim	Biology – Honor Thesis	May 2025-	UNC-Chapel Hill
Yoo Rim Lee	Nutrition - Honor Thesis	March 2025-May 2026	UNC-Chapel Hill
Weston Matthews	Biology	Aug. 2024- May 2025	UNC-Chapel Hill

Nehal Devpura	Nutrition - Honor Thesis	Feb. 2023- May 2026	University of Arizona medical school
Rebecca Zasloff	Biology - Honors Thesis	Aug. 2021 – Jun. 2022	Duke and Harvard school of public health

5) Professional Service

2024-Present President of UNC Microbiome Trainee club

2025-2027 Co-chair of Gordon Research Seminar (GRS): Lactic Acid Bacteria Biology, Symbioses and Applications

2025-Present Leadership of Nutritional Microbiology ASN's Groups Engaging Members (GEMs)

2023-2024 UNC Biological & Biomedical Science Program (BBSP) Leadership Group

2025.8 Judge of iGEM Mini Jamboree in Healthcare category

2023.9-Present Laboratory Animal Coordinator at UNC Microbiome Core

2023-Present Reviewer, Gut Microbiome

2025-Present Reviewer, Food Bioscience

Member since:

2022 International scientific association for probiotics and prebiotics Students and Fellows Association (ISAPP-SFA)

2024 American Society for Nutrition (ASN)

2024 North Carolina Branch of the American Society for Microbiology (NC-ASM)

2024 American Society for Microbiology (ASM)

Awards:

2022 Travel award for International scientific association for probiotics and prebiotics annual conference, Denver

2026 Travel award for International scientific association for probiotics and prebiotics annual conference, Japan